

AI-Driven System for Oil Spill Identification and Monitoring

Leveraging deep learning and satellite imagery for environmental monitoring

Presented by
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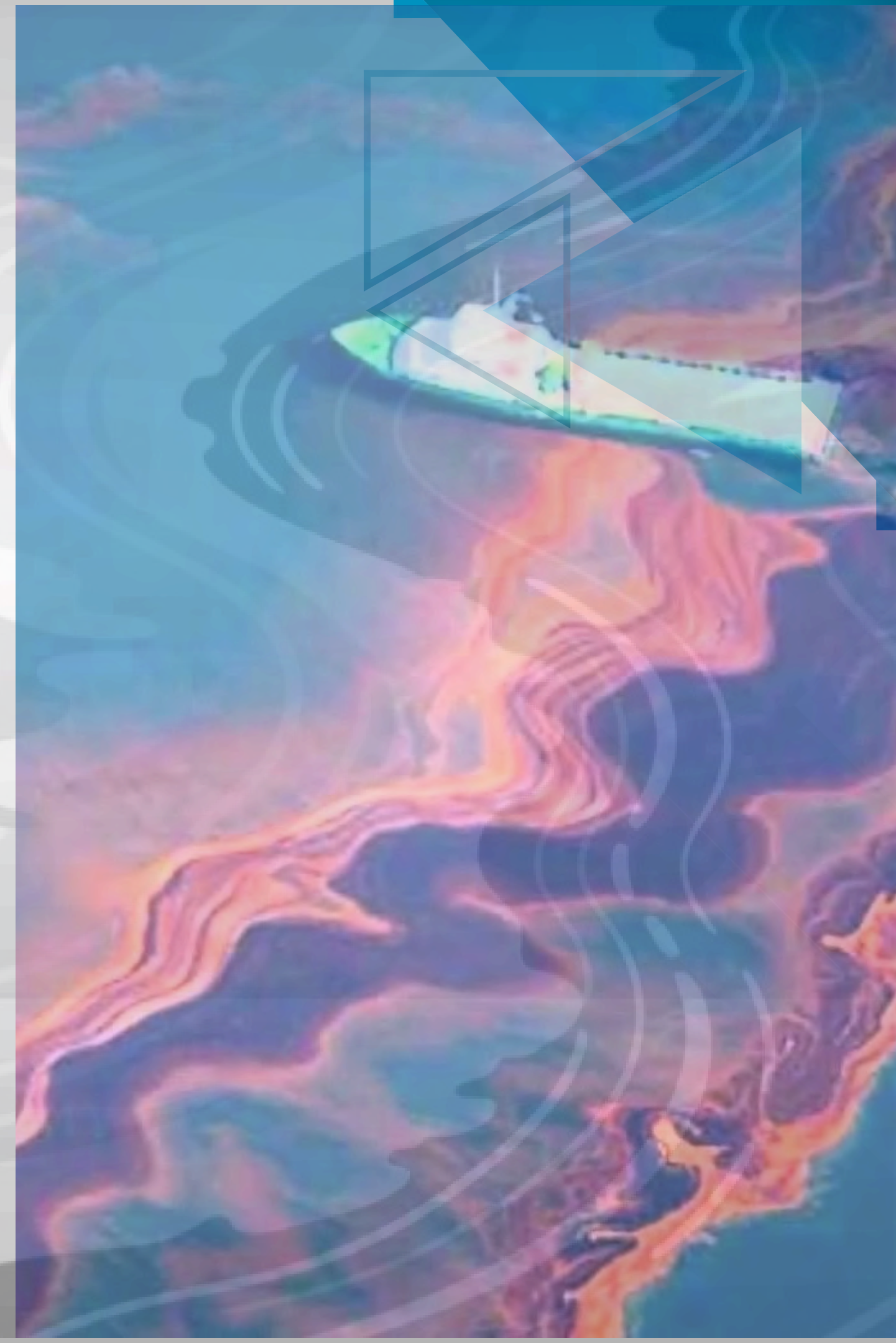




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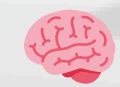
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INTRODUCTION & OVERVIEW

OBJECTIVE

- **Detect and monitor accidental oil spills in oceans using AI-driven automation.**
- **Utilize Synthetic Aperture Radar (SAR) imagery for reliable detection under all weather conditions.**
- **Develop a Deep Learning–based Attention U-Net model for accurate segmentation of oil-affected areas.**
- **Create an intuitive Streamlit web application for real-time visualization and user interaction.**

• OUTCOME

- **Achieved precise segmentation of oil spills from satellite images.**
- **Enabled real-time image analysis and prediction through the Streamlit interface.**
- **Contributed to sustainable marine ecosystem protection using AI technology. Bridged the gap between research and real-world environmental monitoring solutions.**



DATASET OVERVIEW & KEY INSIGHTS

- **Dataset Source: Zenodo (Oil Spill Detection Dataset)**
- **Data Type: Satellite images (SAR) with corresponding segmentation masks**

DATASET ORGANIZATION:

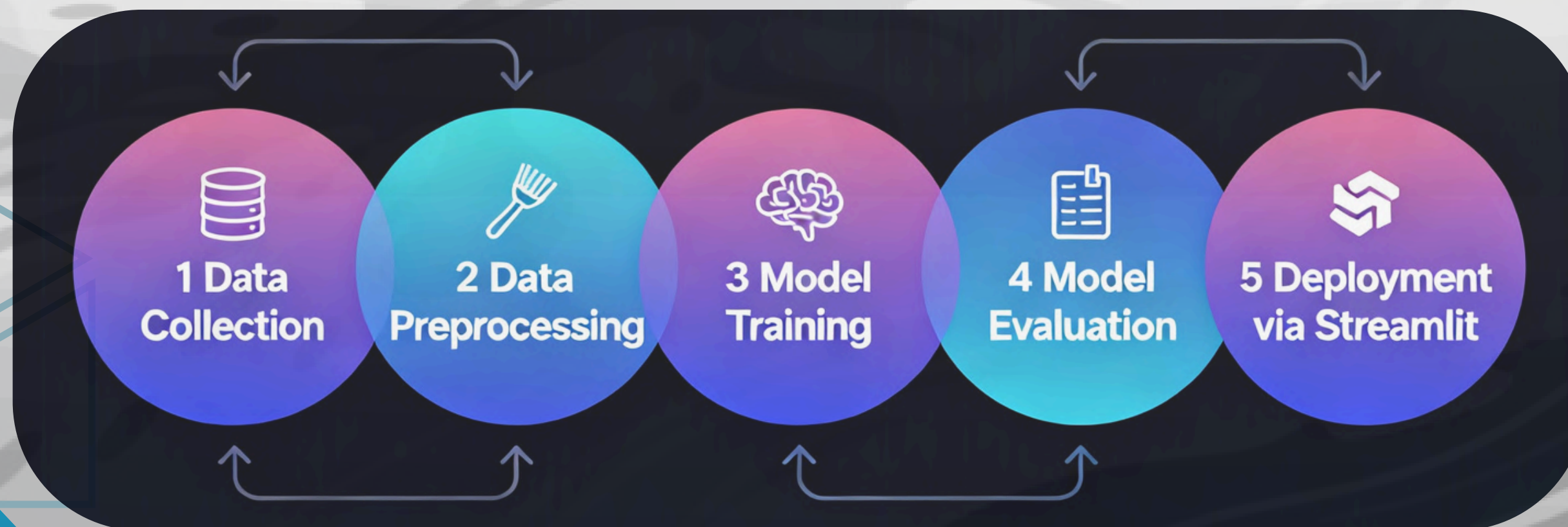
```
/content/drive/MyDrive/Oil_Spill_Detection/dataset 1
/content/drive/MyDrive/Oil_Spill_Detection/dataset/train 0
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/content/drive/MyDrive/Oil_Spill_Detection/dataset/train/images 811
/content/drive/MyDrive/Oil_Spill_Detection/dataset/val 0
/content/drive/MyDrive/Oil_Spill_Detection/dataset/val/masks 203
/content/drive/MyDrive/Oil_Spill_Detection/dataset/val/images 203
/content/drive/MyDrive/Oil_Spill_Detection/dataset/test 0
/content/drive/MyDrive/Oil_Spill_Detection/dataset/test/masks 254
/content/drive/MyDrive/Oil_Spill_Detection/dataset/test/images 254
```

Split	Images	Masks
Train	811	811
Validation	203	203
Test	254	254

- **Images vary in resolution from (768×432) to (5184×3888).**
- **Masks are binary — 1 for oil spill, 0 for background.**
- **Distribution shows class imbalance — spill regions occupy smaller portions.**
- **Sample visualization used to verify alignment of images and masks.**

⚙️ METHODOLOGY

- **Data Collection:** Gathered satellite imagery (SAR) with annotated oil spill regions for model training.
- **Data Exploration & Preprocessing:** Analyzed data distribution, resized images, normalized pixels, and applied noise reduction filters.
- **Model Development:** Designed an Attention-based U-Net for accurate oil spill segmentation.
- **Training & Evaluation:** Trained the model using optimized loss functions and evaluated performance with IoU, Dice, and Precision metrics.
- **Visualization of Results:** Generated comparative visual outputs — original, ground truth, and predicted masks with overlays.
- **Deployment via Streamlit App:** Integrated the trained model into a web interface for real-time image upload, prediction, and analysis.





EXPLORATORY DATA ANALYSIS

- Checked sample SAR images with their corresponding binary masks.
- Verified dataset structure and image dimensions.
- Ensured correct mapping between image and mask pairs.
- Checked data balance between oil spill and non-spill images.
- Analyzed pixel intensity distribution for visual clarity.
- Identified and noted noisy or corrupted images for cleaning.
- Confirmed readiness of data for preprocessing and training.
- **Outcome: Properly paired image–mask dataset prepared for binary segmentation.**





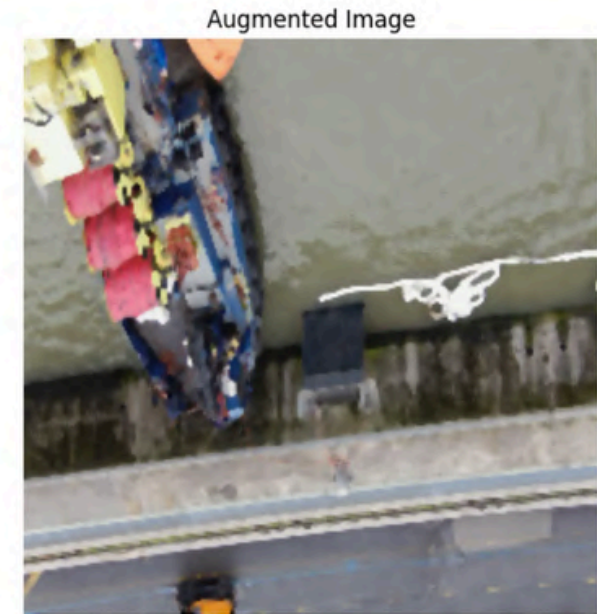
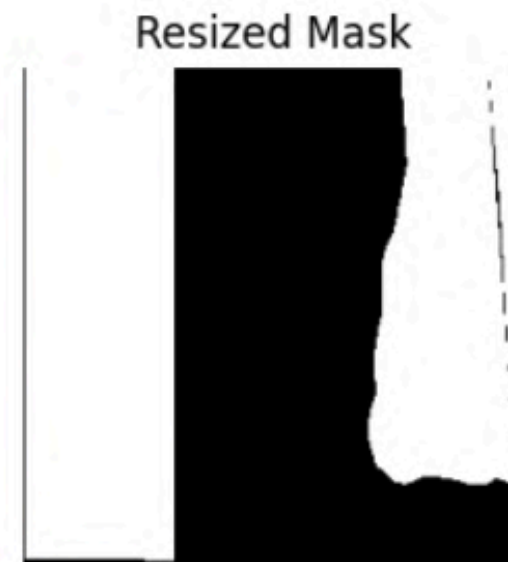
DATA PREPROCESSING

◆ Resizing & Normalization

- Loaded SAR images with their corresponding binary masks.
- Normalized pixel values from [0–255] to [0–1] and resized all images to 256×256.
- Applied Median Filtering to reduce speckle noise while preserving details.
- Verified alignment between each image and its mask after preprocessing.
- 📊 Outcome: Dataset standardized, denoised, and ready for model input.

◆ Data Augmentation

- Applied random flips and rotations to increase data variety.
- Adjusted brightness and contrast to simulate real-world lighting changes.
- Used scaling and cropping to improve generalization during training.
- 🚀 Outcome: Created diverse and robust samples, enhancing model performance and reducing overfitting.



🔍 FEATURE EXTRACTION

◆ PURPOSE:

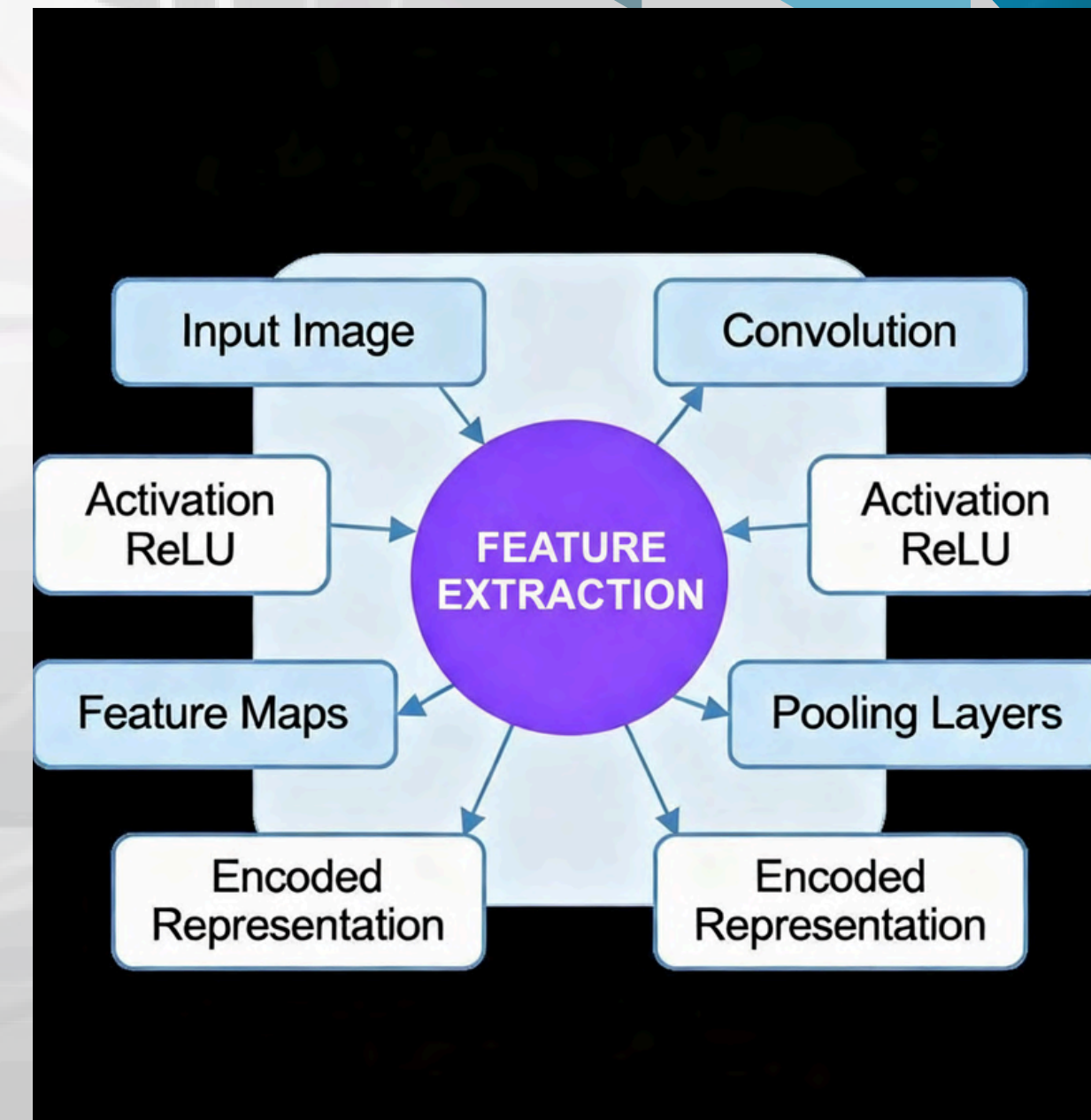
- To automatically extract spatial and textural patterns (features) from SAR images that differentiate oil spill and non-spill regions.

◆ HOW IT WORKS:

- **Encoder Layers:** Capture hierarchical spatial features from low-level textures to high-level shapes.
- **Residual Connections:** Preserve fine-grained edge and boundary details during downsampling.
- **Squeeze-and-Excite (SE) Blocks:** Learn channel-wise importance — focus on key features in SAR intensity maps.
- **Attention Gates:** Filter out irrelevant background noise and highlight oil-affected areas.
- **Bottleneck Layer:** Extracts the most discriminative features for segmentation.

◆ OUTCOME:

- Rich feature maps representing oil textures, shapes, and contrast patterns — ready for mask prediction in the decoder.





MODEL ARCHITECTURE

ATTENTION U-NET – MODEL ARCHITECTURE OVERVIEW

◆ WHAT IS ATTENTION U-NET?

- **An advanced version of the U-Net designed for image segmentation.**
- **It integrates attention gates to focus on vital regions (oil spills) while ignoring irrelevant background (like ocean noise).**

◆ CORE COMPONENTS

ENCODER (CONTRACTING PATH)

- **Extracts key features via convolution + pooling.**
- **Reduces image size while enriching feature depth.**

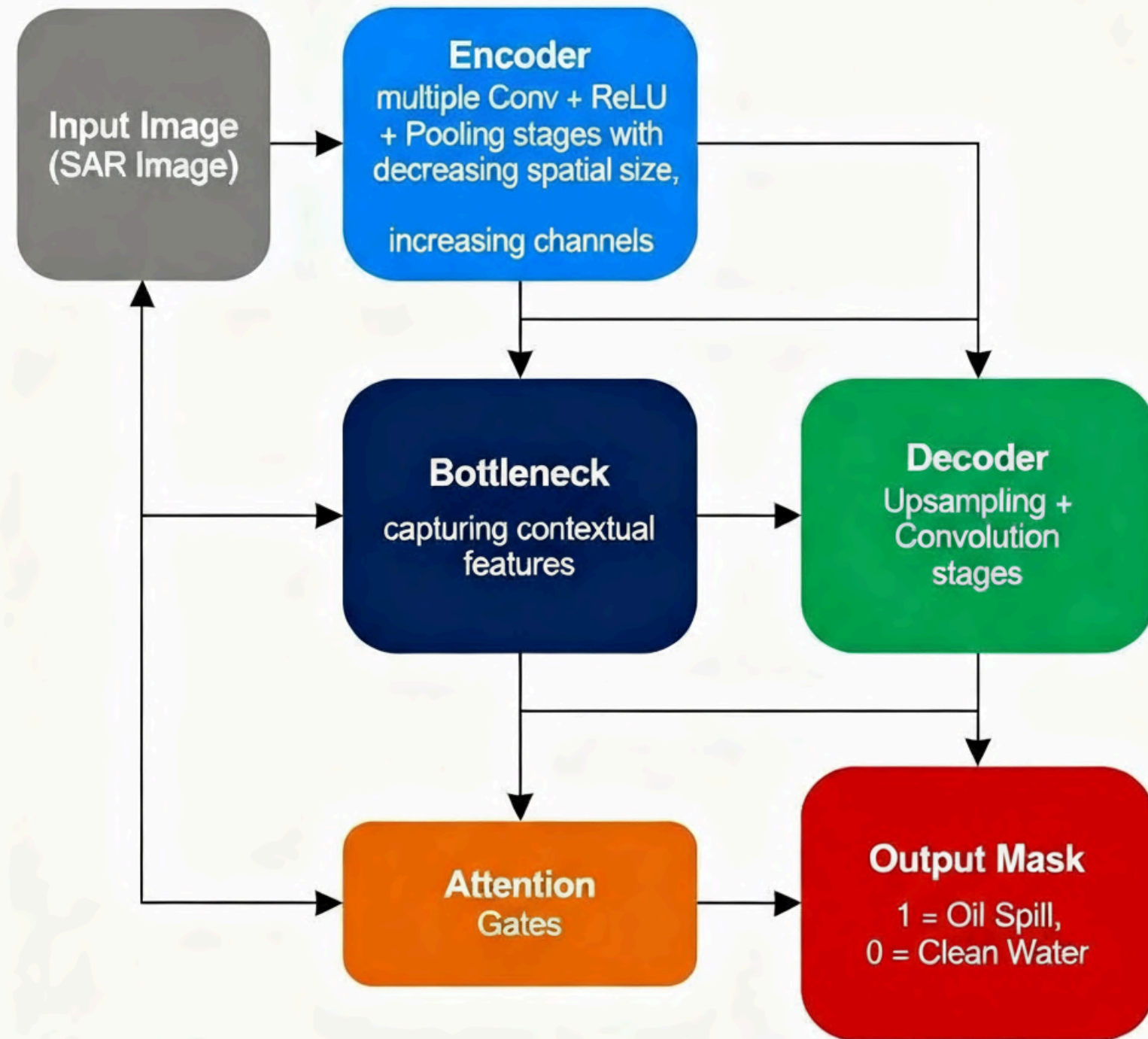
BOTTLENECK LAYER

- **Captures high-level context of oil spill patterns.**

DECODER (EXPANDING PATH)

- **Reconstructs the segmentation mask using upsampling + skip connections.**
- **Skip Connections**
- **Preserve fine spatial details from encoder to decoder.**

Attention U-Net Architecture for Oil Spill Segmentation



◆ SKIP CONNECTIONS

- **Preserve fine spatial details from encoder to decoder.**

◆ ATTENTION GATES

- **Highlight oil-affected regions and suppress noise.**

✚ PURPOSE

- **Achieve precise oil spill segmentation by enhancing feature focus and reducing background interference.**

◆ DATA FLOW

- **Input Image → Encoder → Bottleneck → Decoder → Attention Gates + Skip Connections → Output Mask**

◆ TRAINING PROCESS

- **Loss Function: BCE + Dice Loss (pixel-level accuracy)**
- **Optimizer: Adam (adaptive learning)**
- **Learning Rate Scheduling: Ensures smooth convergence**

◆ MODEL OUTPUT

- **1 → Oil Spill, 0 → Clean Water**
- **Produces sharp boundaries and fewer false detections**



TRAINING & EVALUATION

⚙️ MODEL COMPILATION

- **Framework:** TensorFlow / Keras
 - **Optimizer:** Adam — ensures adaptive learning and smooth convergence.
 - **Loss Function:** Binary Cross-Entropy + Dice Loss → Enhances segmentation accuracy by balancing oil spill vs clean water detection.
 - **Metrics Used:** Dice Coefficient, Accuracy, Precision, Recall, IoU
 - **Learning Rate:** $1e-4$ (automatically reduced when progress slows)
 - **Validation Split:** 15% (for model generalization check)
- 🌱 **Purpose:** Defines how the model learns, what it optimizes, and how performance is measured before starting training.

🧠 MODEL TRAINING

- **Batch Size:** 8
- **Epochs:** 25
- **Training Data:** Preprocessed SAR images (resized, normalized).
- **Data Augmentation:** Random flips, rotations, and contrast adjustments — improve robustness to variations.

⚙️ CALLBACKS USED

- **ModelCheckpoint:** Saves the best-performing model (best_oil_spill_model_v2.h5) based on validation Dice score.
- **EarlyStopping:** Stops training if no improvement for 20 epochs, restoring the best model weights.
- **ReduceLROnPlateau:** Decreases the learning rate by 70% when improvement plateaus.

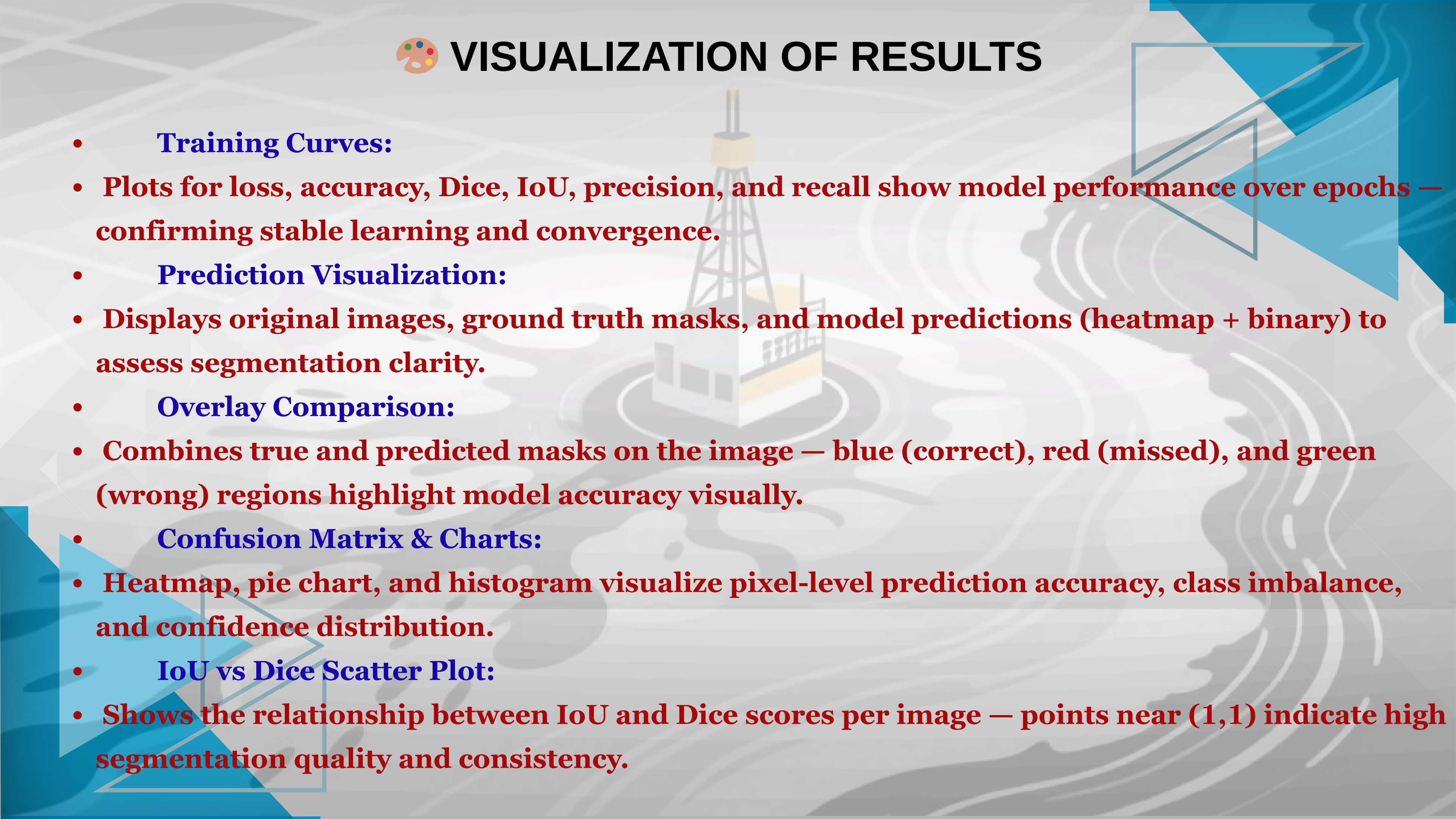
🧠 EVALUATION RESULTS

- **Testing Data:** Unseen SAR satellite images.
- **Metrics Used:**
 - **Dice Coefficient (F1 Score):** Measures overlap accuracy between predicted and actual oil spill areas.
 - **IoU (Intersection over Union):** Evaluates segmentation quality.
 - **Precision / Recall:** Assesses false detections and missed spills.

```
Epoch 23: val_dice_coef_improved did not improve from 0.96772
87/87 ----- 88s 1s/step - accuracy: 0.9620 - dice_coef_improved: 0.9794 - iou_improved: 0.9599 - loss: 0.0462
- precision_improved: 0.9646 - recall_improved: 0.9950 - val_accuracy: 0.9369 - val_dice_coef_improved: 0.9676 - val_iou_impr
oved: 0.9388 - val_loss: 0.0779 - val_precision_improved: 0.9428 - val_recall_improved: 0.9956 - learning_rate: 3.7056e-05
Epoch 24/25
87/87 ----- 0s 911ms/step - accuracy: 0.9588 - dice_coef_improved: 0.9778 - iou_improved: 0.9569 - loss: 0.050
3 - precision_improved: 0.9608 - recall_improved: 0.9958
Epoch 24: val_dice_coef_improved did not improve from 0.96772
87/87 ----- 89s 1s/step - accuracy: 0.9589 - dice_coef_improved: 0.9779 - iou_improved: 0.9570 - loss: 0.0503
- precision_improved: 0.9608 - recall_improved: 0.9958 - val_accuracy: 0.9370 - val_dice_coef_improved: 0.9677 - val_iou_impr
oved: 0.9389 - val_loss: 0.0782 - val_precision_improved: 0.9427 - val_recall_improved: 0.9958 - learning_rate: 3.1739e-05
Epoch 25/25
87/87 ----- 0s 863ms/step - accuracy: 0.9628 - dice_coef_improved: 0.9800 - iou_improved: 0.9611 - loss: 0.045
5 - precision_improved: 0.9653 - recall_improved: 0.9955
Epoch 25: val_dice_coef_improved did not improve from 0.96772
87/87 ----- 86s 985ms/step - accuracy: 0.9628 - dice_coef_improved: 0.9800 - iou_improved: 0.9611 - loss: 0.04
55 - precision_improved: 0.9653 - recall_improved: 0.9955 - val_accuracy: 0.9370 - val_dice_coef_improved: 0.9677 - val_iou_i
mproved: 0.9390 - val_loss: 0.0778 - val_precision_improved: 0.9428 - val_recall_improved: 0.9958 - learning_rate: 2.7357e-05
Restoring model weights from the end of the best epoch: 20.
```

✅ Training complete!

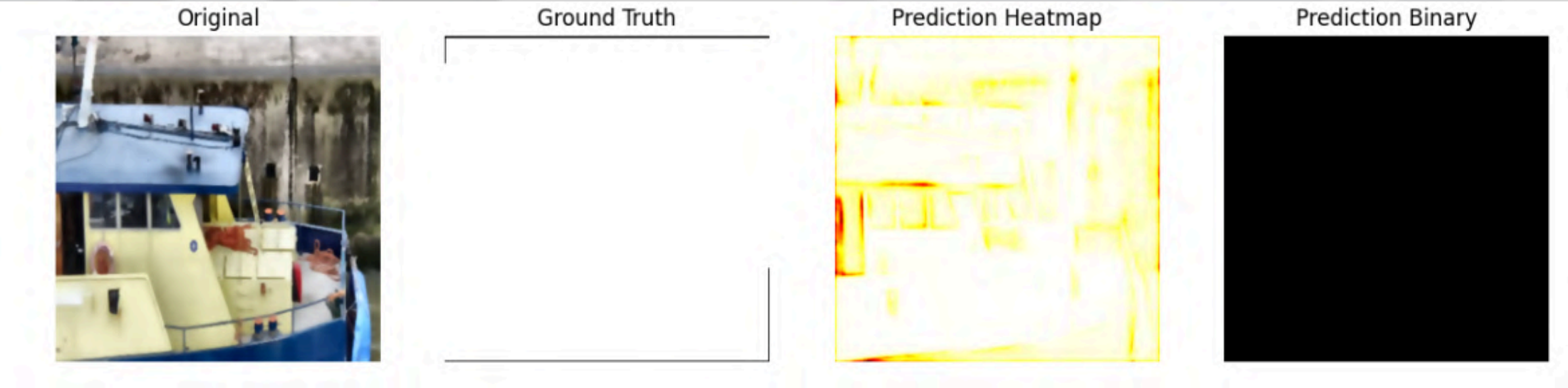
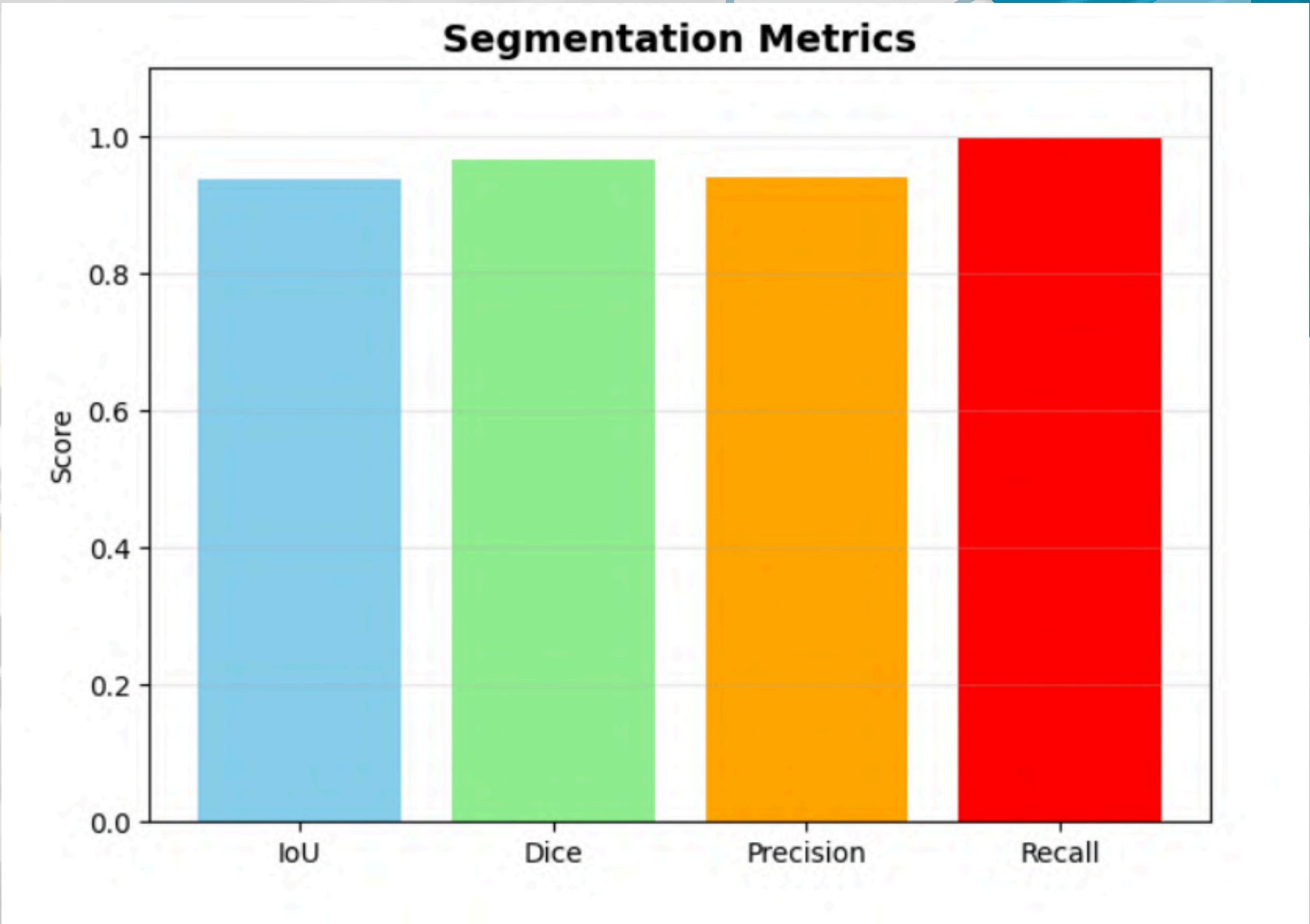
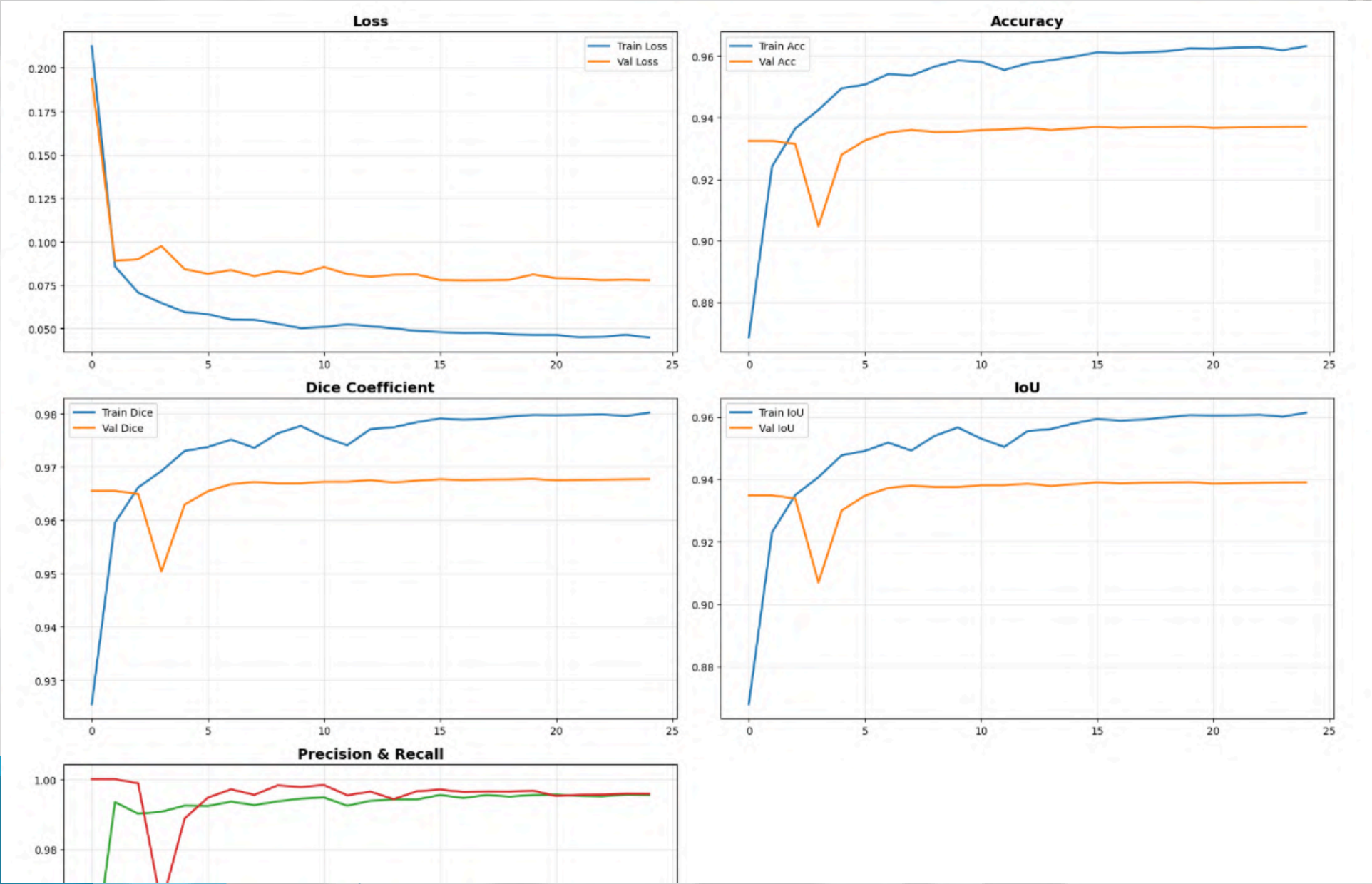
✅ Validation Metrics:	
BCE Loss	: 0.2514
Dice Loss	: 0.0323
Accuracy	: 0.9371
Dice Coef	: 0.9677
IoU	: 0.9390
Precision	: 0.9395
Recall	: 0.9967



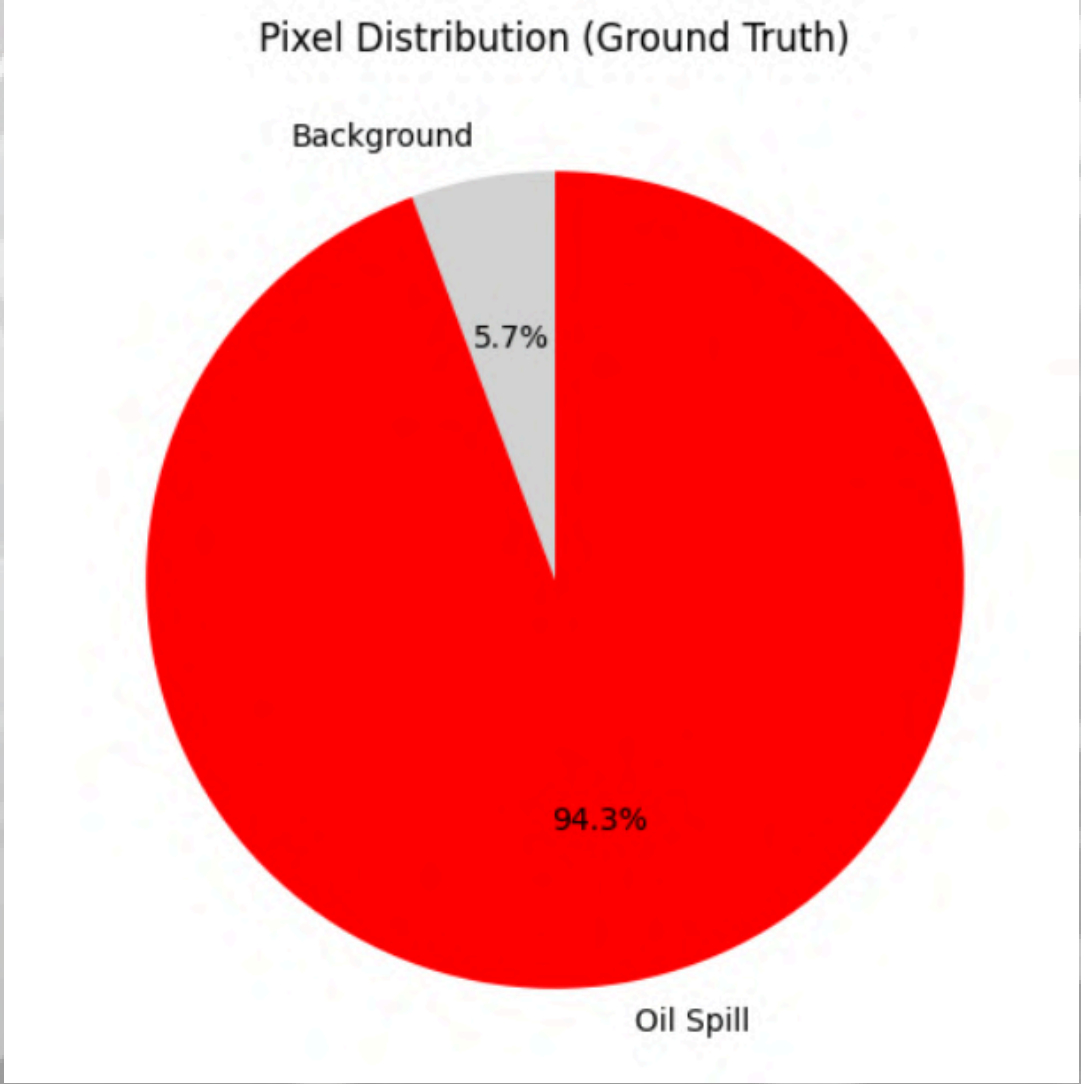
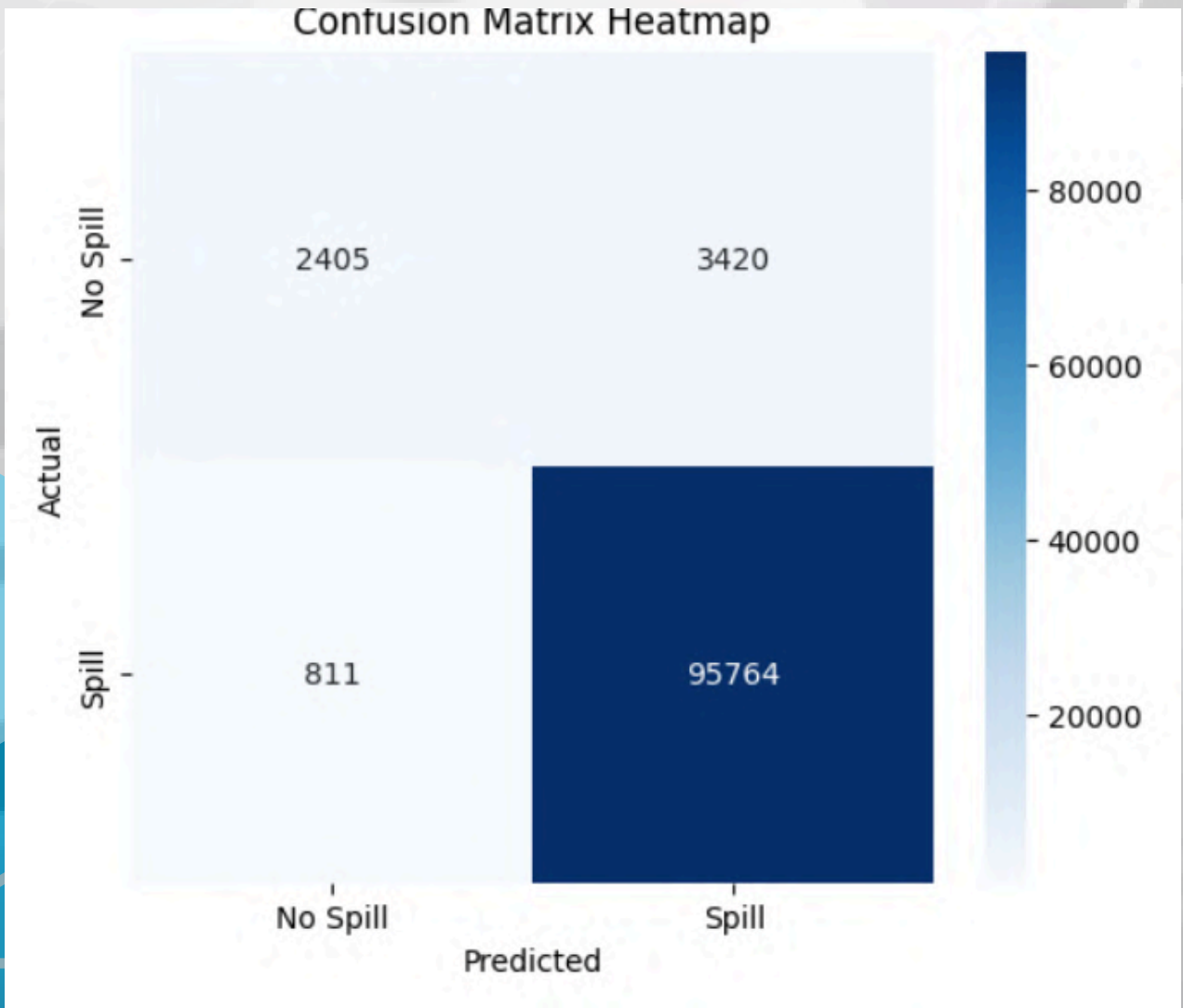
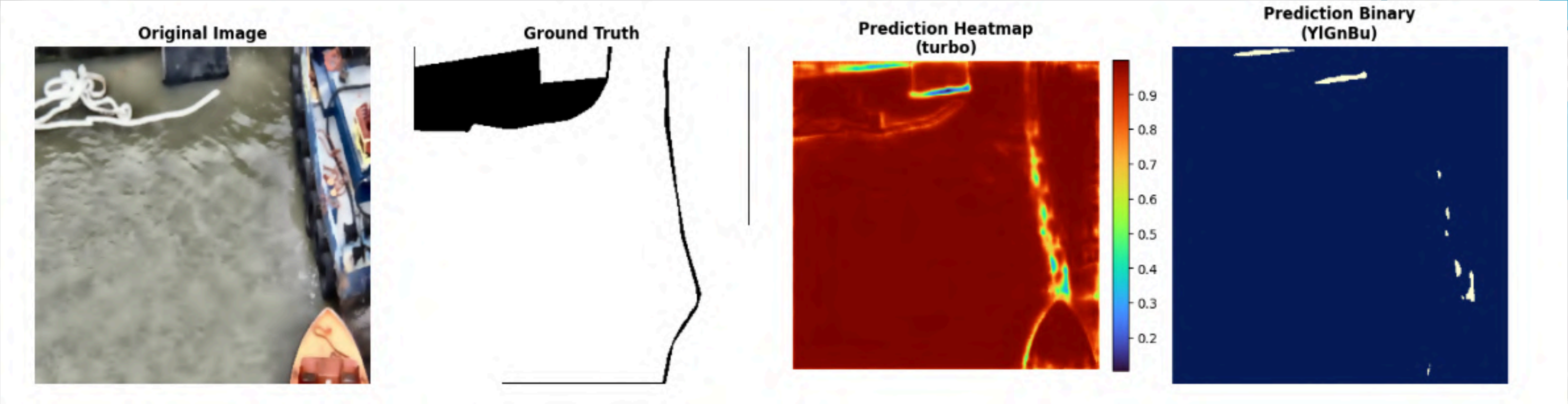
VISUALIZATION OF RESULTS

- **Training Curves:**
- **Plots for loss, accuracy, Dice, IoU, precision, and recall show model performance over epochs — confirming stable learning and convergence.**
- **Prediction Visualization:**
- **Displays original images, ground truth masks, and model predictions (heatmap + binary) to assess segmentation clarity.**
- **Overlay Comparison:**
- **Combines true and predicted masks on the image — blue (correct), red (missed), and green (wrong) regions highlight model accuracy visually.**
- **Confusion Matrix & Charts:**
- **Heatmap, pie chart, and histogram visualize pixel-level prediction accuracy, class imbalance, and confidence distribution.**
- **IoU vs Dice Scatter Plot:**
- **Shows the relationship between IoU and Dice scores per image — points near (1,1) indicate high segmentation quality and consistency.**

RESULTS



RESULTS





STREAMLIT WEB INTERFACE



DEPLOYMENT OVERVIEW

- **The Oil Spill Detection System** is a Streamlit-based web application that allows users to upload satellite images and automatically detect oil spills using a pre-trained deep learning model.
- It integrates SQLite for managing user data, detection history, and feedback records — making it a lightweight, standalone deployable system.

DEPLOYMENT STEPS

- **Model Loading:** The trained AI model (best_oil_spill_model_v2.h5) is loaded at startup.
- **App Launch:** The Streamlit framework serves the web interface.
- **User Interaction:** Users log in, upload images, and view results.
- **Database Operations:** Each detection, feedback, and user registration is stored in SQLite.
- **Admin Panel:** Admins can monitor users, feedback, and detection records.



STREAMLIT WEB INTERFACE

Layer	Technologies Used	Purpose
Frontend	Streamlit	Web UI and interaction
Backend	Python	Application logic
Database	SQLite	User, history, and feedback storage
AI Model	Keras (TensorFlow backend)	Oil spill segmentation
Visualization	Matplotlib, Plotly	Graphs, charts, overlays
Other Tools	NumPy, OpenCV, PIL	Image preprocessing
Version Control	Git & GitHub	Source management
Deployment	Streamlit Cloud / Local host	Running and testing the app

★ KEY FEATURES

-  **User Authentication** – Secure login and registration with SQLite database.
-  **AI-Based Detection** – Upload satellite or aerial images; the trained model detects and highlights oil spills.
-  **Detection History** – Saves all past detections with image name, confidence score, severity, and timestamp.
-  **Feedback System** – Users can send their feedback through an integrated feedback form.
-  **Admin Panel** – View all registered users, detection logs, and feedback entries.

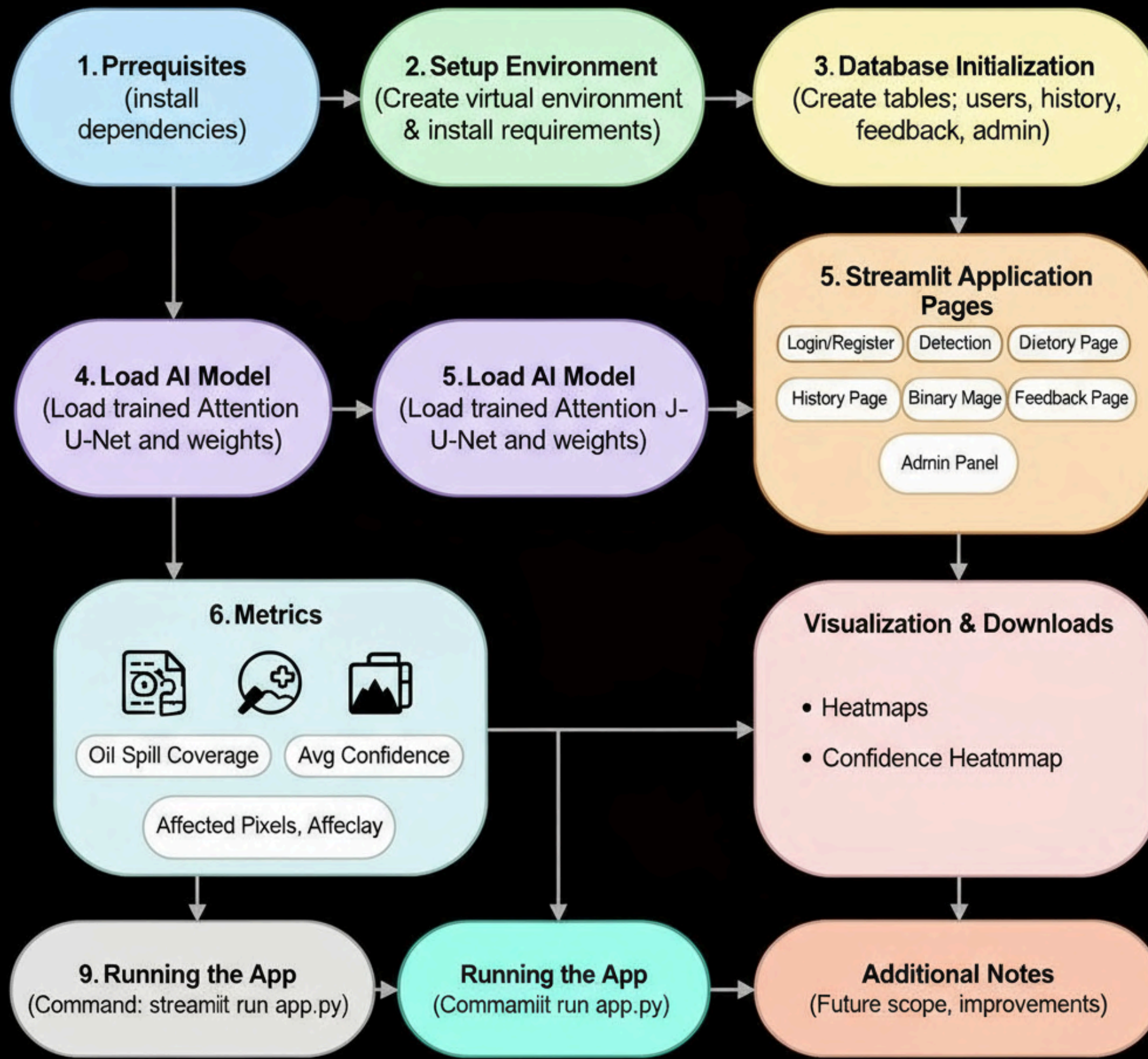
 **Visualization Dashboard** – Displays performance metrics along with result overlays and charts.

 **Download Option** – Allows users or admins to download segmented images for documentation.

 **User Experience Enhancements** – Includes email links, pop-up balloons, and success alerts for better interactivity.

DEPLOYMENT WORKFLOW

Oil Spill Detection App – Deployment Workflow



DEPLOYMENT RESULTS

The screenshot displays the user interface of the Oil Spill Detection System. At the top, the header reads "Oil Spill Detection System" with the tagline "AI-powered detection, monitoring & analysis". Below this, the main heading "Oil Spill Detection System" is repeated. The interface includes a "Login" button and a "Register" button. A section titled "Login to Your Account" contains input fields for "Username" and "Password", followed by a large orange "LOGIN" button. Below the login section, there is a text box with the placeholder "Advanced AI-powered oil spill detection from satellite imagery". The footer of the application states "Oil Spill Detection Project", "Developed by Madhu Mitha | Infosys SpringBoard Virtual Internship", and "© 2025 All Rights Reserved".



Download Results

-  [DOWNLOAD ALL VISUALIZATIONS](#)
-  [DOWNLOAD BINARY MASK](#)
-  [DOWNLOAD RED OVERLAY](#)
-  [DOWNLOAD PROBABILITY HEATMAP](#)



DEPLOYMENT RESULTS

 **Detection History** ↗

BACK

User: Madhu

Total detections: 4


Oil (1233).jpg

Date: 2025-10-31 01:34:07

Coverage

94.65%

Confidence

98.8%

Area

9.693 km²

Severity

 **CRITICAL**

Deploy ⋮

 **ction System**
Monitoring & analysis

HISTORY

FEEDBACK

LOGOUT

ADMIN PANEL

 **Oil Spill Detection System**
AI-powered detection, monitoring & analysis

Send Feedback

BACK

User: Madhu

 We'd love to hear from you!

Your Feedback

HI HOW ARE U

Your email (optional)

GENERATE EMAIL LINK

[Click here to open your email client and send feedback](#)

 Your feedback is saved and is ready to send via your email client!



DEPLOYMENT RESULTS

Choose view

Users

Oil Spill Detection System

AI-powered detection, monitoring & analysis

Admin Panel

Registered Users

ID: 5 | Username: intern3 | Email: [i23@gmail.com](#) | Registered On: 2025-10-24 14:07:37

ID: 4 | Username: intern1 | Email: [abc2@gmail.com](#) | Registered On: 2025-10-22 11:42:37

ID: 3 | Username: intern | Email: [abc@gmail.com](#) | Registered On: 2025-10-22 11:39:45

ID: 2 | Username: admin | Email: [i23@gmail.com](#) | Registered On: 2025-10-22 11:36:21

Choose view

Detection History

Oil Spill Detection System

AI-powered detection, monitoring & analysis

Admin Panel

Detection History

Filter by username (optional)

ID: 10 | User: Madhu | File: Oil (1233).jpg | Coverage: 94.65% | Confidence: 98.8% | Severity: CRITICAL | Area: 9.693 km² | Time: 2025-10-31 01:34:07

ID: 9 | User: intern3 | File: Oil (576).jpg | Coverage: 91.48% | Confidence: 98.8% | Severity: CRITICAL | Area: 9.368 km² | Time: 2025-10-24 14:10:01

ID: 8 | User: intern3 | File: Oil (576).jpg | Coverage: 91.48% | Confidence: 98.8% | Severity: CRITICAL | Area: 9.368 km² | Time: 2025-10-24 14:09:18

ID: 7 | User: intern1 | File: Oil (142).jpg | Coverage: 91.37% | Confidence: 98.7% | Severity: CRITICAL | Area: 9.356 km² | Time: 2025-10-22 11:44:11

ID: 6 | User: intern1 | File: Oil_2 (9).jpg | Coverage: 95.11% | Confidence: 98.7% | Severity: CRITICAL | Area: 9.739 km² | Time: 2025-10-22 11:43:12

ID: 5 | User: intern | File: Oil_2 (3).jpg | Coverage: 97.09% | Confidence: 98.7% | Severity: CRITICAL | Area: 9.942 km² | Time: 2025-10-22 11:41:21

ID: 4 | User: intern | File: Oil (210).jpg | Coverage: 92.29% | Confidence: 98.6% | Severity: CRITICAL | Area: 9.451 km² | Time: 2025-10-22 11:40:17

Choose view

Feedback

Oil Spill Detection System

AI-powered detection, monitoring & analysis

Admin Panel

User Feedback

ID: 6 | User ID: 1 | Username: Madhu | Time: 2025-10-31 01:38:24

Feedback: HI HOW ARE U

CHALLENGES FACED

Model Optimization:

- Fine-tuning the segmentation model to achieve high Dice and IoU scores without overfitting required multiple training iterations.

Image Preprocessing Issues:

- Handling different image resolutions, noise levels, and contrast variations from satellite data was challenging during input standardization.

Database Integration:

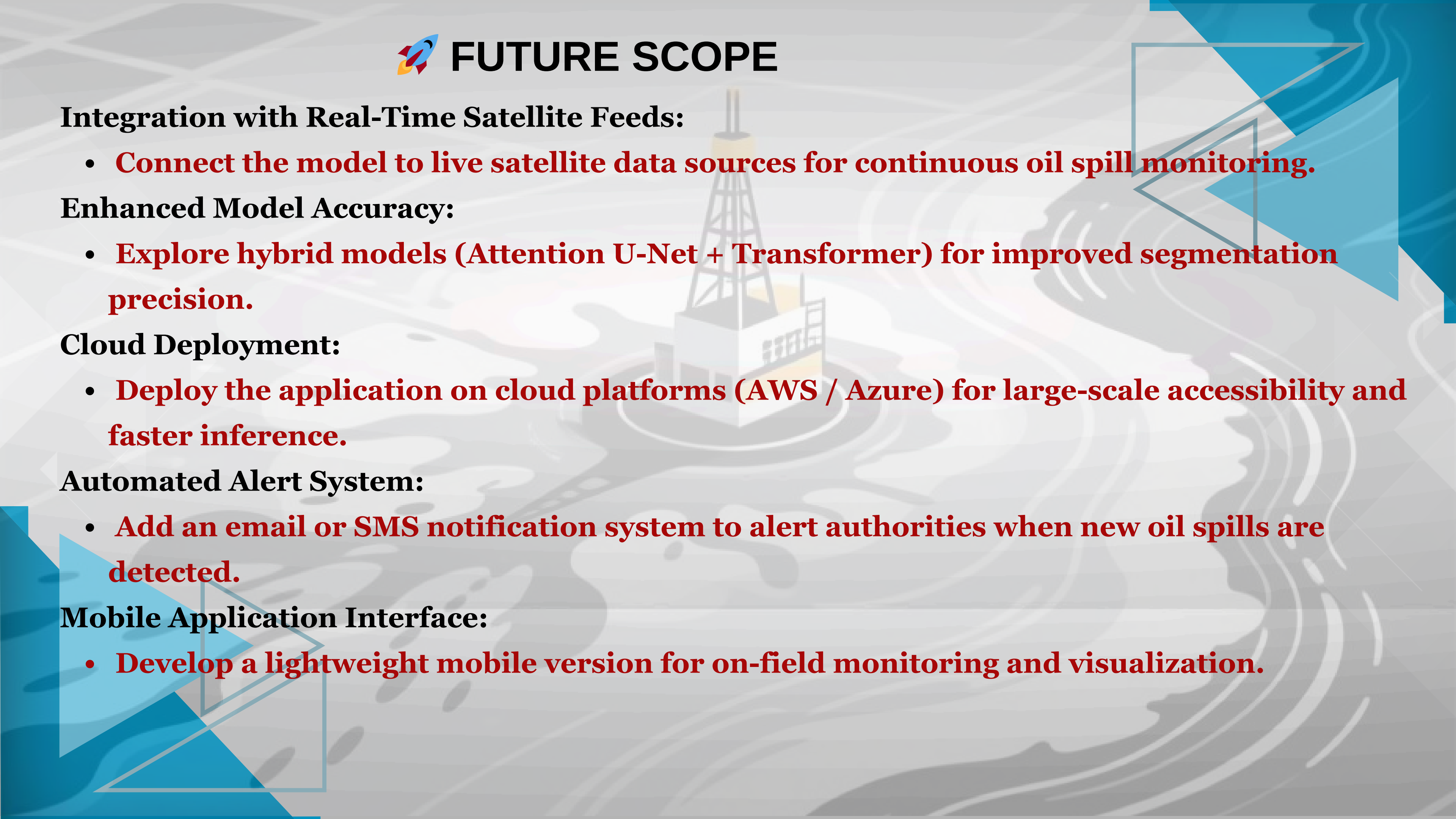
- Linking the detection module with SQLite to store user activity, feedback, and history while maintaining performance and data consistency.

Visualization Complexity:

- Designing intuitive and informative visualizations (Dice vs IoU plots, overlays, histograms) that clearly interpret the model's performance.

Real-Time Inference Performance:

- Managing prediction latency while loading large images and generating segmentation masks efficiently on limited hardware.



FUTURE SCOPE

Integration with Real-Time Satellite Feeds:

- **Connect the model to live satellite data sources for continuous oil spill monitoring.**

Enhanced Model Accuracy:

- **Explore hybrid models (Attention U-Net + Transformer) for improved segmentation precision.**

Cloud Deployment:

- **Deploy the application on cloud platforms (AWS / Azure) for large-scale accessibility and faster inference.**

Automated Alert System:

- **Add an email or SMS notification system to alert authorities when new oil spills are detected.**

Mobile Application Interface:

- **Develop a lightweight mobile version for on-field monitoring and visualization.**



CONCLUSION

The Oil Spill Detection System demonstrates the effective use of deep learning in addressing real-world environmental challenges. Using the Attention U-Net model, it accurately detects and segments oil spill regions from satellite images while minimizing background interference.

Integrated with a Streamlit-based web interface, the system enables users to upload images, visualize detection results, and review key performance metrics. Supported by an SQLite database, it ensures secure data handling for users, predictions, and feedback.

This project not only enhances the accuracy of spill identification but also promotes the adoption of AI for environmental sustainability. Its scalable design and user-friendly interface make it suitable for future integration with real-time satellite data and alert systems, contributing to faster response and improved marine safety.

The background features a stylized illustration of an oil rig on a body of water, rendered in a muted, desaturated color palette. Overlaid on this are several geometric elements: a large, light blue triangle in the top right corner, a smaller blue triangle in the bottom left corner, and a series of concentric, wavy lines in shades of blue and grey that emanate from the rig, suggesting ripples or a signal. The text "THANK YOU" is centered in a bold, blue, serif font.

THANK YOU